Chapter 16

Matrices

a spreadsheet you can do sums with

	bread	milk	eggs
North shop	12	30	6
South shop	8	25	10



drop the labels, keep the grid

A **matrix** is a table of numbers in rows and columns, like a spreadsheet with the labels rubbed out. The North shop sold 12 loaves, 30 milks and 6 boxes of eggs; the South shop sold 8, 25 and 10. Keeping the numbers in their grid positions lets you do arithmetic on the whole table at once.

Your module says matrix **methods** are not assessed, but **solving simultaneous equations** is, and Lecture 12 solves them with matrices. So this chapter builds up just enough to follow that: reading a matrix, adding, multiplying (the only odd one), and the inverse, which is an undo button.

Goes with **Lecture 12** (matrices and systems of equations).

Reading a matrix: rows, columns, and seat numbers

Rows run across (like rows of seats in a cinema). **Columns** run up and down (like the columns holding up a building). The **size** is always said **rows first, then columns**:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

2 × 2
2 rows, 2 columns

$$\begin{bmatrix} 12 & 30 & 6 \\ 8 & 25 & 10 \end{bmatrix}$$

row 1 highlighted

2 × 3
2 rows, 3 columns

$$\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$$

3 × 2
3 rows, 2 columns

Seat numbers. Each number in a matrix is an **entry** (or element). Its address is written with two small numbers below the letter, a **subscript**: a_{ij} means the entry in **row i, column j**, like "row C, seat 7".

$$\begin{bmatrix} 12 & 30 & 6 \\ 8 & 25 & 10 \end{bmatrix}$$

a_{23} = 10: go down to row 2, then across to column 3.

a_{12} = 30 (row 1, column 2)

Memory hook: "RC", like a remote-controlled car: Rows before Columns, for sizes and for addresses.

Adding, subtracting and scaling: cell by cell

RULE: adding, subtracting, scaling and equal matrices

Add or subtract only matrices of the **same size**: combine the entries in the same position. Different sizes: no answer.

Scalar multiplication kA: multiply **every** entry by the number k (a whole number, a negative or a fraction).

Equal matrices: $A = B$ means the same size and **every** pair of matching entries equal. So one matrix equation gives one ordinary equation per entry.

Week 1 and week 2 sales from the two shops. Total sales? Add each cell to the **matching cell**:

$$\begin{bmatrix} 12 & 30 & 6 \\ 8 & 25 & 10 \end{bmatrix} + \begin{bmatrix} 10 & 28 & 4 \\ 9 & 20 & 7 \end{bmatrix} = \begin{bmatrix} 22 & 58 & 10 \\ 17 & 45 & 17 \end{bmatrix}$$

12 + 10 = 22, 30 + 28 = 58, …

Lecture 12's picture of this: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} \text{pizza} & \text{kebab} \\ \text{curry} & \text{soup} \end{bmatrix}$ puts a + pizza top-left, and so on. Things only combine with the thing in the **same seat**. So both matrices must be the **same size**; a 2 × 3 plus a 2 × 2 has no answer.

$$\begin{bmatrix} 2 & 3 & -1 \\ 7 & 0 & 4 \end{bmatrix} - \begin{bmatrix} -3 & 4 & 2 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & -1 & -3 \\ 6 & 0 & 2 \end{bmatrix}$$

the slide's subtraction: 2 − (−3) = 5

Scaling (multiplying by one plain number, called a **scalar**): every cell gets multiplied. Four weeks at the same rate:

$$4 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix}$$ ← *4 × every entry*

A fraction works the same way. Multiplying by $\frac{1}{3}$ is dividing every entry by 3:

$$\frac{1}{3} \begin{bmatrix} 3 & 6 \\ -9 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -3 & \frac{1}{3} \end{bmatrix}$$ ← *every entry ÷ 3; leave 1/3 as a fraction*

Equal matrices. If $\begin{bmatrix} x+1 & 4 \\ 3 & 2y \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 3 & 10 \end{bmatrix}$, match the entries seat by seat:

$$x+1=5 \Rightarrow x=4, \qquad 2y=10 \Rightarrow y=5$$

top-left and bottom-right seats

Multiplying: row meets column

RULE: multiplying two matrices

The entry in **row i, column j** of AB = **row i of A** paired with **column j of B**: multiply first with first, second with second, ..., then **add**.

Do this for every row of A with every column of B. It is **not** cell-by-cell.

Multiplying two matrices is **not** cell by cell. It comes from a shopping bill. You buy 2 coffees and 3 cakes; a coffee costs £2.50 and a cake £2. Write the order as a **row** and the prices as a **column**:

order $\begin{bmatrix} 2 & 3 \end{bmatrix}$ prices $\begin{bmatrix} 2.5 \\ 2 \end{bmatrix} = 2 \times 2.5 + 3 \times 2 = 11$

first with first, second with second, then add: the bill is £11

That's the whole rule: pair up, multiply, add. For bigger matrices you just do it for every row of the first with every column of the second. Lecture 12's example, traced with two highlighters (yellow row, blue column):

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \to \begin{bmatrix} 19 & \\ & \end{bmatrix}$

row 1 × column 1: 1×5 + 2×7 = 19

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \to \begin{bmatrix} 19 & 22 \\ & \end{bmatrix}$

row 1 × column 2: 1×6 + 2×8 = 22

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \to \begin{bmatrix} 19 & 22 \\ 43 & \end{bmatrix}$

row 2 × column 1: 3×5 + 4×7 = 43

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \to \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$

row 2 × column 2: 3×6 + 4×8 = 50

Slide check: Lecture 12 writes the bottom-left entry as $3 \times 5 + 4 \times 8 = 47$. Row 2 of A is (3, 4) and column 1 of B is (5, 7), so it's $3 \times 5 + 4 \times 7 = 15 + 28 = 43$. The full answer is $\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$.

Which sizes can multiply? Dominoes

RULE: the size rule

$(m \times n)(n \times p) = (m \times p)$.

AB exists only when the **number of columns of A** equals the **number of rows of B** (the two inside numbers match). The answer has the rows of A and the columns of B (the outside numbers).

Check the sizes **before** multiplying. AB can exist while BA doesn't, and even when both exist, $AB \neq BA$ in general.

Row-meets-column only works if a row of the first is **as long as** a column of the second (every item needs a price). Write each size as a domino and put them side by side: the **touching numbers must match**, and the **outside numbers give the answer's size**.



→ **2 × 1**



→ **1 × 1 (a single number, like the bill)**



→ **3 ≠ 2: can't be done**

Lecture 12's "different size" example: $\begin{bmatrix} 1 & 4 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ is (2 × 2)(2 × 1), so the answer is 2 × 1:

$$\begin{bmatrix} 1 \times 2 + 4 \times 5 \\ 6 \times 2 + 3 \times 5 \end{bmatrix} = \begin{bmatrix} 22 \\ 27 \end{bmatrix}$$

← *each row meets the one column*

Swap them round and it fails: $\begin{bmatrix}2\\5\end{bmatrix}\begin{bmatrix}1&4\\6&3\end{bmatrix}$ is (2 × 1)(2 × 2); the inside numbers 1 and 2 don't match, so this product **does not exist**. Same two matrices, one order works and the other doesn't.

$$(2\times 3)(3\times 4)=2\times 4, \qquad (3\times 4)(2\times 3):\ 4\neq 2,\ \text{undefined}$$

Order matters: CD is not DC

RULE: order of multiplication

In general $AB\neq BA$: matrix multiplication is **not commutative**. Never swap the order of a product to make it easier.

"Multiply A by B on the left" means BA; "on the right" means AB.

With ordinary numbers $3\times 5=5\times 3$. With matrices, swapping the order usually changes the answer. Lecture 12's "give it a try":

$$C=\begin{bmatrix}7&8\\2&-1\end{bmatrix}, \quad D=\begin{bmatrix}3&2\\-1&4\end{bmatrix}$$

$$CD=\begin{bmatrix}7(3)+8(-1)&7(2)+8(4)\\2(3)+(-1)(-1)&2(2)+(-1)(4)\end{bmatrix}=\begin{bmatrix}13&46\\7&0\end{bmatrix}$$

$$DC=\begin{bmatrix}3(7)+2(2)&3(8)+2(-1)\\-1(7)+4(2)&-1(8)+4(-1)\end{bmatrix}=\begin{bmatrix}25&22\\1&-12\end{bmatrix}$$

Completely different! Think of socks and shoes: socks-then-shoes is not the same as shoes-then-socks. (The slide's heading says "calculate CD" again, but it's working out DC.)

The matrix that acts like 1

RULE: the identity matrix

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (1s on the leading diagonal, 0s elsewhere).

$AI = IA = A$ for any square matrix A of the same size: I behaves like the number 1.

A **diagonal** matrix has numbers only on the diagonal from top-left to bottom-right (zeros elsewhere). The **identity** matrix I is the diagonal matrix with every diagonal entry 1:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$ ← *unchanged!*

Multiplying by I changes nothing, just like multiplying a number by 1. Check the top-left entry: row (1, 2) meets column (1, 0): $1 \times 1 + 2 \times 0 = 1$. The zeros make sure each entry only "sees" itself.

Simultaneous equations as one matrix sum

Here's why matrices are in a course about equations. Multiply a 2 × 2 by a column of letters (Lecture 12):

$$\begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x + 2y \\ 5x + 3y \end{bmatrix}$$ ← *row meets column, with letters*

Those are the left-hand sides of two equations! So the pair $x + 2y = 13$ and $2x - 5y = 8$ can be written as **one** matrix equation:

$$\begin{bmatrix} 1 & 2 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 13 \\ 8 \end{bmatrix} \qquad AX = B$$

A = numbers in front *X = unknowns* *B = right-hand sides*

It looks just like $3x = 12$, one equation with one unknown. To solve $3x = 12$ you'd undo the "× 3". So we need an **undo button for a matrix**.

Honest note: for two equations, the Chapter 15 methods are quicker by hand. The matrix way wins for computers and for big systems, like Lecture 12's joke equation with 12 unknowns (which needs 12 independent equations).

The inverse: an undo button (2 × 2 recipe)

RULE: determinant and inverse of a 2 × 2 matrix

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ the **determinant** is $\det A = ad - bc$.

If $\det A = 0$ the matrix is **singular**: it has **no inverse**.

If $\det A \neq 0$: $A^{-1} = \dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (swap a and d, change the signs of b and c, divide by the determinant).

The inverse satisfies $AA^{-1} = A^{-1}A = I$. Multiply out to check an inverse.

Only square matrices can have inverses.

For numbers, the undo for "× 3" is "× $\frac{1}{3}$" because $\frac{1}{3} \times 3 = 1$. For a matrix A, the undo is written A^{-1} ("A inverse") and it satisfies $A^{-1}A = I$. Lecture 12 gives the recipe for a 2 × 2:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

1. swap the two diagonal entries (a and d) **2.** change the sign of the other two **3.** divide everything by $ad - bc$

The number $ad - bc$ is called the **determinant**. If it's 0 you'd divide by zero, so there's **no inverse**. That happens exactly when the two equations aren't independent (Chapter 15: the same line, or parallel lines).

PROCEDURE: solving AX = B

1. Write the equations as $AX = B$ (A = coefficients, X = unknowns, B = right-hand sides).
2. Find $\det A$. If it is 0, stop: no unique solution.
3. Find A^{-1} with the recipe.
4. $X = A^{-1}B$, with A^{-1} on the **left**: multiply both sides on the left by A^{-1}, and $A^{-1}AX = IX = X$. (**Not** BA^{-1}: order matters, and here it doesn't even exist.)
5. Check the answers in the original equations.

Solve $AX = B$ with $A = \begin{bmatrix} 1 & 2 \\ 2 & -5 \end{bmatrix}$. Tiny steps:

$$ad - bc = (1)(-5) - (2)(2) = -9$$ ← the determinant (not 0: good)

$$A^{-1} = \frac{1}{-9}\begin{bmatrix} -5 & -2 \\ -2 & 1 \end{bmatrix}$$ ← swap 1 and −5; flip signs of the 2s

$$X = A^{-1}B$$ ← undo A on both sides (inverse × A = I, and IX = X)

$$= \frac{1}{-9}\begin{bmatrix} (-5)(13) + (-2)(8) \\ (-2)(13) + (1)(8) \end{bmatrix} = \frac{1}{-9}\begin{bmatrix} -81 \\ -18 \end{bmatrix}$$ ← row meets column

$$= \begin{bmatrix} 9 \\ 2 \end{bmatrix} \Rightarrow x = 9,\ y = 2$$ ← ÷ (−9)

Check in the original equations: $9 + 2(2) = 13$ ✓ and $2(9) - 5(2) = 8$ ✓.

You can't divide by a matrix. $X = \frac{B}{A}$ means nothing. The "undo" is multiplying by A^{-1}, and because order matters you must say which side. AX has A on the left of X, so A^{-1} goes on the left too. Here B is 2 × 1 and A^{-1} is 2 × 2, so BA^{-1} is (2 × 1)(2 × 2): it doesn't exist.

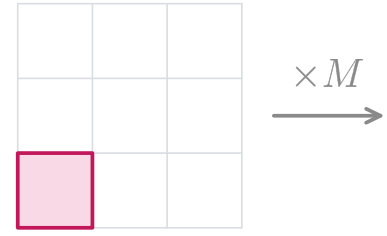
Check an inverse: multiply it by the original and you must get I. Is $\begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}$ the inverse of $\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$?

$$\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 4-3 & -12+12 \\ 1-1 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

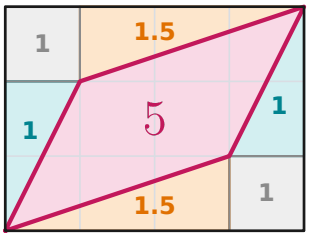
= I, so yes

What the determinant measures: area

$ad - bc$ looks like a random recipe, but it has a shape. Multiply a matrix by the corner points of a 1 by 1 square, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and they land on the matrix's two **columns**. For $M = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ the square gets pushed over into a slanted shape (a parallelogram):



area 1



the box around it: 4 × 3 = 12

Its area? Take the 4 × 3 box (12) and cut off the corner pieces: 1.5 + 1.5 + 1 + 1 + 1 + 1 = 7.

12 − 7 = 5, and ad − bc = 3×2 − 1×1 = 5 too!

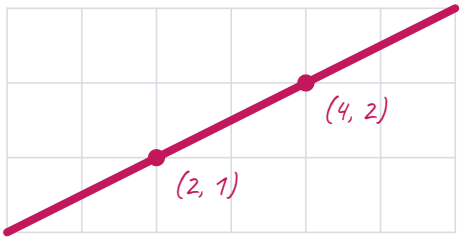
That always happens. With letters the box is $(a+b)$ by $(c+d)$, the orange triangles make one $a \times c$ rectangle, the teal ones one $b \times d$, and there are two grey $b \times c$ rectangles:

$$(a+b)(c+d) - ac - bd - 2bc = ad - bc$$

the box, minus the corners

So the **determinant is how much the matrix stretches area**: every shape comes out 5 times bigger under M. A minus sign means the square also got **flipped over**, like a mirror image: the inverse example's determinant of -9 means "9 times the area, and flipped".

And why 0 means no undo button. Try $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$: $ad - bc = 4 - 4 = 0$. Its columns point the same way, so the square gets squashed completely flat:

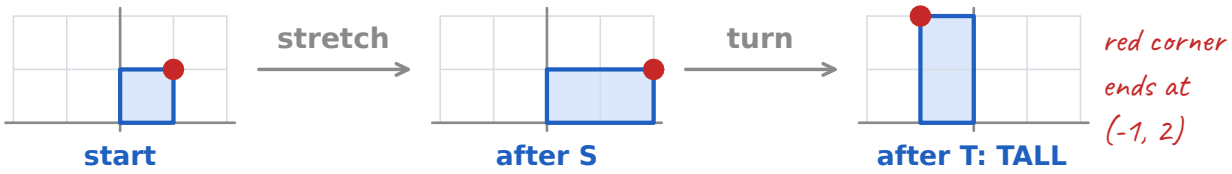


The whole square lands on one line: area 0. Many different points land on the same spot, so there's no way to tell where each came from. You can't un-squash a line back into a square, so there is no inverse. That's the "divide by zero" warning, as a picture.

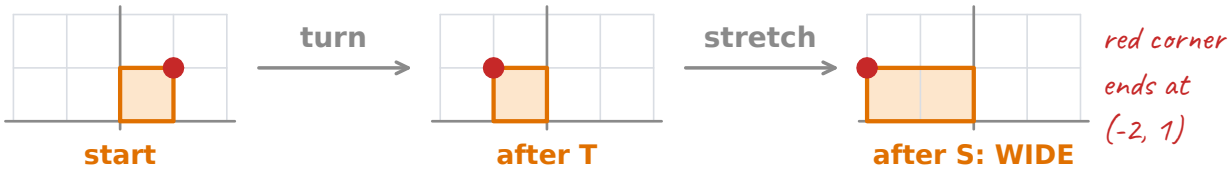
Order matters, seen: stretch then turn, or turn then stretch?

A matrix moves every point, so two matrices in a row are two moves in a row. Take $S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ (**stretch** sideways ×2) and $T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (**turn** a quarter turn anticlockwise). Follow the square and its **red corner**:

stretch first, then turn



turn first, then stretch



Same two moves, different order, different result: one ends up tall, the other wide. **Watch the order of the letters:** "S first, then T" is written TS, because the matrix nearest the point acts first ($TS\,\mathbf{v} = T(S\mathbf{v})$). That's exactly why $CD \neq DC$.

✎ YOUR GO (pencil first)

To UNDO "stretch, then turn" (get the tall shape back to the start square), which move do you undo first: the stretch or the turn?

Answer: undo the **last** move first. Turn back, then un-stretch, just as you take off your shoes before your socks. So the inverse reverses the order:

$$(TS)^{-1} = S^{-1}T^{-1}$$ ← *undo the turn first, then the stretch*

Find the unknown: missing entries

PROCEDURE: partial-information matrix problems

1. Turn each fact into ordinary equations. "No inverse" or "singular" means $ad - bc = 0$. "$AB = C$" means multiply out AB and match it entry by entry with C. "B is the inverse of A" means $AB = I$.
2. Solve those equations for the unknowns.
3. Substitute back and check (work out the determinant or the product again).

Example: find k so that $\begin{bmatrix} k & 4 \\ 3 & 6 \end{bmatrix}$ has no inverse.

$$\det = 6k - 4 \times 3 = 6k - 12 \quad \leftarrow \text{ad – bc}$$

$$6k - 12 = 0 \;\Rightarrow\; k = 2 \quad \leftarrow \text{no inverse: determinant 0}$$

Check: $\begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$: $12 - 12 = 0$ ✓. Its second row is 1.5 times the first: the two equations it stands for are not independent.

Example: for which values of k is $\begin{bmatrix} k & 2 \\ 8 & k \end{bmatrix}$ singular?

$$k \times k - 2 \times 8 = k^2 - 16 = 0 \quad \leftarrow \text{ad – bc = 0}$$

$$k^2 = 16 \;\Rightarrow\; k = \pm 4 \quad \leftarrow \text{both square roots count}$$

Example: $\begin{bmatrix} a & 1 \\ 2 & b \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$. Find a and b.

$$\begin{bmatrix} 3a + 1 \\ 6 + b \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix} \quad \leftarrow \text{multiply out the left side}$$

$$3a + 1 = 7 \Rightarrow a = 2, \qquad 6 + b = 10 \Rightarrow b = 4$$

match the entries

Example: $\begin{bmatrix} 2 & p \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 4 & 1 \\ 1 & q \end{bmatrix} = \begin{bmatrix} 11 & 8 \\ 7 & 7 \end{bmatrix}$. Find p and q.

row 1, col 1: $8 + p = 11 \;\Rightarrow\; p = 3$

row 1, col 2: $2 + pq = 8 \;\Rightarrow\; 2 + 3q = 8 \;\Rightarrow\; q = 2$

use p = 3

row 2: $4 + 3 = 7, \quad 1 + 3q = 1 + 6 = 7$ ← *check the other two entries ✓*

Example: $\begin{bmatrix} 2 & -1 \\ k & 3 \end{bmatrix}$ is the inverse of $\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$. Find k.

row 2 of A with col 1: $5(2) + 2k = 0$ ← *AB = I, so the bottom-left entry is 0*

$10 + 2k = 0 \;\Rightarrow\; k = -5$

Check: with $k = -5$ the whole product is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ✓.

Mistakes to avoid

✗ Common mistakes

✗ $\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}\begin{bmatrix}5 & 6\\7 & 8\end{bmatrix} = \begin{bmatrix}5 & 12\\21 & 32\end{bmatrix}$ ✓ $= \begin{bmatrix}19 & 22\\43 & 50\end{bmatrix}$

multiplying is row meets column, not cell by cell

✗ $AB = BA$ ✓ $AB \neq BA$

order matters; never swap a product round

✗ $X = \dfrac{B}{A}$ ✓ $X = A^{-1}B$

there is no dividing by a matrix: multiply by the inverse

✗ $X = BA^{-1}$ ✓ $X = A^{-1}B$

the inverse goes on the same side as A (the left)

✗ $\det = ad + bc$ ✓ $\det = ad - bc$

it is a subtraction; and the inverse is not 1 over each entry

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Size	$m \times n$ rows × columns, rows first (also for addresses: a_{ij} is row i, column j)
Add / subtract	same size only; combine entries in the same position
Scalar multiple	$k\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$ every entry, including fractions like $\frac{1}{3}$
Equal matrices	same size and every matching entry equal: one equation per entry
Multiplying	entry (i, j) of AB = row i of A with column j of B: multiply pairs, add
Size rule	$(m \times n)(n \times p) = m \times p$ inside numbers must match, or the product doesn't exist
Order	$AB \neq BA$ in general; AB can exist when BA doesn't
Identity	$AI = IA = A$ I has 1s on the diagonal, 0s elsewhere
Determinant	$\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$ 0 means **singular**: no inverse
Inverse	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ check with $AA^{-1} = A^{-1}A = I$
Solving AX = B	$X = A^{-1}B$ A^{-1} on the left; never divide by a matrix
Undoing two moves	$(TS)^{-1} = S^{-1}T^{-1}$ undo the last move first

Spreadsheets: your turn

Answers in grey. Trace every multiplication with two fingers: one along the row, one down the column.

Reading

1. Size of [1 2 3]?
2. Size of a matrix with 4 rows and 2 columns?
3. In [[5, 1], [7, 9]], what is a_{21}?

1. 1 × 3 2. 4 × 2 3. 7

Adding and scaling

4. [[1, 2], [3, 4]] + [[0, 5], [−1, 2]]
5. 3 × [[2, −1], [0, 4]]
6. [[6, 3], [2, 1]] − [[1, 1], [1, 1]]

4. [[1, 7], [2, 6]] 5. [[6, −3], [0, 12]] 6. [[5, 2], [1, 0]]

Multiplying

7. [1 4] × [[2], [3]]
8. [[2, 1], [0, 3]] × [[1], [4]]
9. [[1, 2], [0, 1]] × [[3, 0], [1, 2]]
10. Can a 3 × 2 multiply a 3 × 2?

7. [14] 8. [[6], [12]] 9. [[5, 4], [1, 2]] 10. No: inner numbers 2 and 3 don't match

Inverse and equations

11. Determinant of [[4, 3], [2, 1]]
12. Inverse of [[2, 1], [5, 3]]
13. Solve 2x + y = 5, 5x + 3y = 13 using your answer to 12

11. 4 − 6 = −2 12. [[3, −1], [−5, 2]] (determinant 1) 13. x = 2, y = 1

Find the unknown

14. Find k so that [[k, 6], [2, 4]] has no inverse.
15. For which values of k is [[k, 3], [3, k]] singular?
16. [[2, a], [b, 1]] × [[1], [3]] = [[11], [7]]. Find a and b.
17. [[x + 2, 6], [1, 3y]] = [[5, 6], [1, −6]]. Find x and y.
18. [[2, −3], [−1, k]] is the inverse of [[k, 3], [1, 2]]. Find k.
19. Is [[1, −2], [−1, 3]] the inverse of [[3, 2], [1, 1]]? Show it.

14. k = 3 (4k − 12 = 0) 15. k = ±3 ($k^2 - 9 = 0$) 16. a = 3, b = 4 (2 + 3a = 11, b + 3 = 7) 17. x = 3, y = −2 (match entries) 18. k = 2 (top-left of the product: 2k − 3 = 1) 19. yes: the product is [[1, 0], [0, 1]] = I